MIDTERM · PROJECT STAGE 1

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Polyclinic No. 5 Booking System

How our system connects

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Information and Communication Technologies

One phone line for 27,000 patients

PROBLEM

A busy phone line and a paper appointment book.

Double bookings, lost appointments, no warning when a doctor is absent.

MAIN USERS

Patients of the polyclinic.

Limitation: many are elderly, with poor eyesight and little app experience.

So: large text, few steps, KZ/RU interface.

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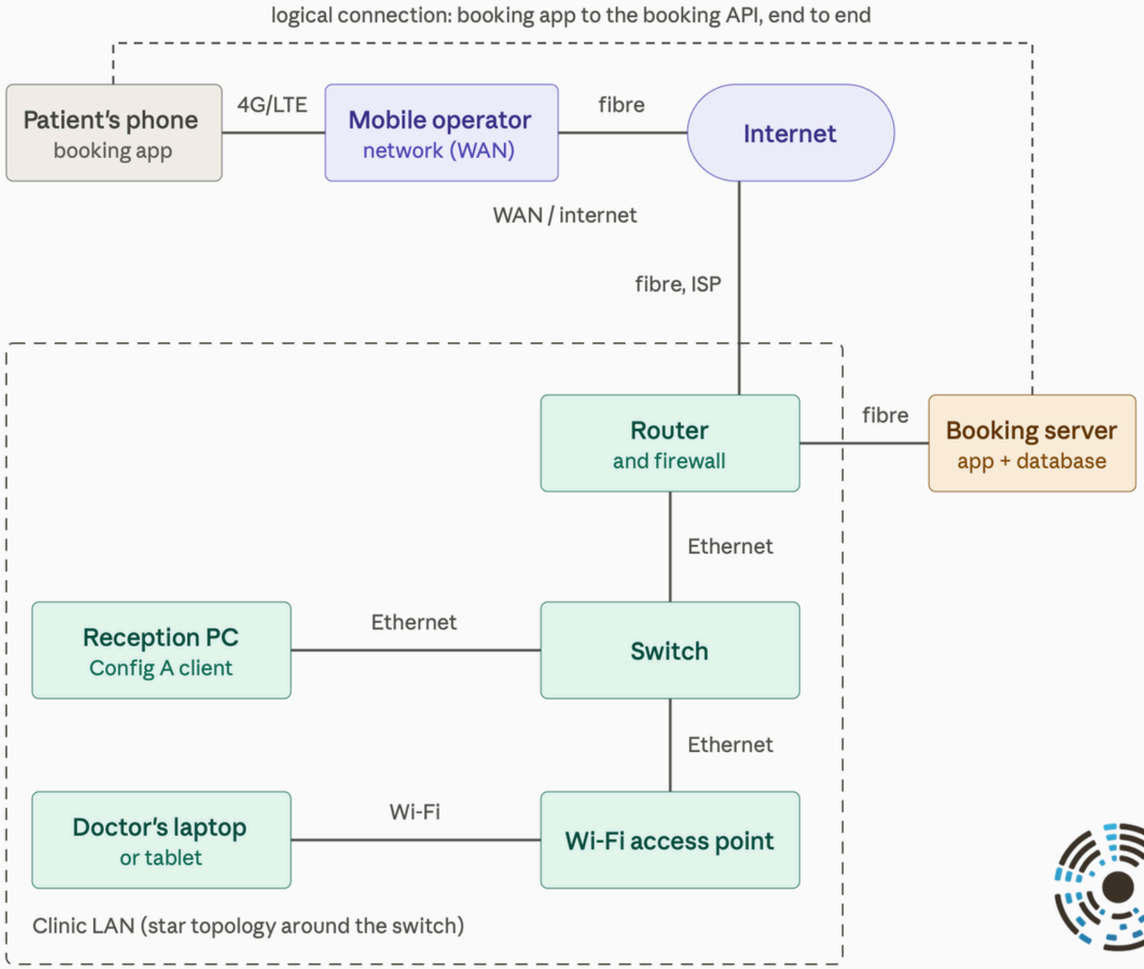
From the phone to the server

SCALE

Phone: PAN
Clinic: LAN
Mobile network and internet: WAN

TOPOLOGY

Star around the switch: one cable per device, easy to extend.



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One tap through five layers

Layer	What happens to the booking	Protocol	Distance
Application	Browser sends it as JSON	HTTPS	Process to process
Transport	Numbered segments, lost ones resent	TCP :443	Process to process
Network	Addressed phone IP to server IP	IP	Host to host
Data	LTE frame, then Ethernet frames	LTE, Ethernet	Node to node
Physical	Radio, light, electric pulses	4G, fibre, copper	Node to node

Why TCP: every field must arrive, in order. Domain: booking.pc5-astana.kz (.kz)

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Where each tier runs

Tier	Runs there	Where	Model	Provider
1 Presentation	Booking app in browser	Phone or reception PC	None	None
2 Application	Booking API	Kazakhstan cloud	PaaS	SmartCloud.kz
3 Data	Database	Megahost, Astana	IaaS	Megahost.kz

Personal data is stored in Astana.
Law No. 94-V, Article 12 is met.

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When the connection drops

1

The screen says "Connection lost, trying again". Doctor, slot and reason stay saved.

2

After reconnecting, the app asks the server if the booking already exists. No duplicates.

3

New flowchart path: "Connected?" after step 6. No loops back to step 6.

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What changed since A1 and A2

KEPT

Three-tier client-server design. The defense asked about microservices; for one clinic with 27,000 patients, three tiers are simpler and cheaper.

ADDED

- Hosting in Kazakhstan: SmartCloud.kz and Megahost.kz, for Law 94-V
- Third error path: connection lost
- Network design: 4G, Ethernet, Wi-Fi
- Domain: booking.pc5-astana.kz

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